Samarth Sharma

Final Year Undergraduate, IIT Kanpur
Major: Electrical Engineering
Minors: Industrial Management Engineering, Literature

samarthsb23@iitk.ac.in
Samarth Sharma
Google Scholar
samarthsb23

Year	Qualification	School/Institution	CPI/%
2023-2027	B.Tech.	Indian Institute of Technology Kanpur	**8.6**/10
2023	CBSE XII	Hope Hall Foundation School (HHFS), Delhi	96.8%
2021	CBSE X	Delhi Public School, Prayagraj	95.8%

Scholastic Achievements

- Served as a **Reviewer** for International Conference on Signal Processing and Communications (SPCOM) 2026 at IISc. Bangalore
- Secured **All India Rank 1082** in JEE Advanced'23 & **All India Rank 954** in JEE Mains'23 among 1 million candidates

Work Experience

Google, Bengaluru

Hardware Engineering Intern May'26 - Jul'26

RTL Design, CI2, Google Cloud

- Deployed an **intelligent** RTL-end **automated waiver generation** and root-cause analysis framework, reducing turnaround time from over **300 hours to under 8 hours** by limiting RTL source tracing to unresolved registers
- Integrated Synopsys' **Implementation Design Check (IDC)** tool to Google's internal design and PD workflow for high-impact **TPU networking** projects **left-shifting** flop analysis saving significant crunch time
- Designed a **Gemini-powered interactive multi-agent** LLM framework for **automated waiver generation**, leveraging IDC's built-in register-RTL mapping reports to limit manual RTL source tracing while enabling **natural-language** waiver specification and designer-guided validation of grouped optimization patterns
- **Benchmarked** IDC against existing optimization and analysis methodologies, evaluating register violation correlation, designer productivity, and waiver coverage with respect to **Fusion Compiler** (FC) while collaborating with **international RTL, Physical Design** teams to identify and report flow gaps to Synopsys

Research Experience

Deep Learning Aided Radar Signal Processing: Algorithmic Augmentation and FPGA Acceleration

Research Intern, Prof. Sumit J. Darak May'25 - Apr'26

Algorithms to Architecture (A2A) lab, IIIT Delhi

- Co-authored **two international conference** papers and **one journal** manuscript in ISAC and RSP domains
- Developed low-latency custom pipelined **HLS IPs** and integrated **PYNQ** drivers using **AXI4** interfaces for **deep-learning-aided** radar signal processing (RSP) algorithms and architectures on **Zynq-MPSoCs** for Integrated Sensing and Communication (**ISAC**) applications
- Created a **state-of-the-art** Quantization Aware Fine Tuning (QAFT) architecture using **INT8** weights to reduce BRAM utilization by **81.9**%
- Designed a novel weight-reconfigurable architecture, providing an **additional 56% performance gain**
- Achieved **59.8%** lower RMSE, and **20.4**× lower latency using novel E2E MLP based architectures over classical algorithms, while using just **20 slow-time packets** to match the **170** packet performance at lower SNRs
- Created a new **OOR** (out-of-range) metric to better capitulate robustness of the proposed architecture while using **Integrated Logic Analyzers** to debug AXI transactions

Relevant Courses

p : PG − level, ∗ : Excellent, † : Online

VLSI System Design^p	Digital Electronics
Microelectronics-II*	Chip based VLSI Design†
Spin-Electronics Devices^{*p}	Data Structures and Algorithms
Communication Systems*	Machine Learning†
Complex Variables*	Linear Algebra, Probability & Statistics

Research Publications

- **International Conference on Signal Processing and Communication (SPCOM), Indian Institute of Science, Bangalore:** *Low-Latency Deep Learning Based Doppler Velocity Estimation for Integrated Sensing and Communication*, **First Author**, July 2026 [IEEE Xplore]
- **IEEE Radar Conference, Phoenix, AZ, USA:** *Reconfigurable Low-Complexity Architecture for High Resolution Doppler Velocity Estimation in Integrated Sensing and Communication System*, May 2026 [arXiv]

Key Projects

RTL design of a Mixed-Signal Backend Controller

EE619 Course Project, Prof. Chithra Jan'26 - Apr'26

- Designed a Verilog **backend controller** for a mixed signal IC with **FSM**-based startup sequencing and serial communication
- Implemented **CDC-safe** multi-clock design and verified functionality using a comprehensive testbench and **GTKWave**
- Performed synthesis and timing analysis, evaluating timing constraints using open-source tools like **Yosys** and **OpenSTA**

Transistor-Level Duty Cycle Correction Circuit on Cadence Virtuoso

Summer Project, Prof. Chithra May'25 - Jun'25

- Designed a **transistor-level** duty cycle correction circuit in **gpdk-180nm** CMOS technology on **Cadence Virtuoso**
- Achieved a corrected duty cycle of **48 to 52%** for **0.5 to 1 GHz** inputs, starting from initial distortions ranging between **30 to 70%**

Semiconductor Device Modeling using Devsim TCAD

Prof. Rituraj Dec'24 - Feb'25

- Explored **DEVSIM TCAD** framework, including device setup, **meshing**, and model definition using Python scripting
- Simulated 1D **p-n junction diode** using **drift-diffusion** equations; analyzed **IV behavior** and **carrier** dynamics
- Integrated solver mechanics, **convergence** behavior via **Newtons method**, and visualized results using **Matplotlib**

DIGIWARE - Digital Design using Verilog

Electrical Engineers' Association, IIT Kanpur Dec'23 - Jan'24

- Explored and optimized various Verilog modules for efficient **RTL design and digital logic synthesis**
- Modeled and verified **combinational** and **sequential** logic, like **FSMs** and **counters**, in **Verilog HDL** and **GTKWave**
- Designed and implemented **multiple real-world digital systems** using **Verilog HDL**, including a smart car parking system, a customized alarm clock and a traffic light controller
- Designed and simulated complex **digital systems** like a car parking system, alarm clock, and traffic controller in Verilog

Technical Skills

- **Languages:** Verilog HDL, SystemVerilog, C, C++, Python, LaTeX
- **Softwares:** Xilinx Vitis HLS, Vivado, Synopsys VC Spyglass, Cadence Virtuoso, Yosys, OpenSTA, MATLAB, MicroCap, GNU Octave, Devsim TCAD, Altium